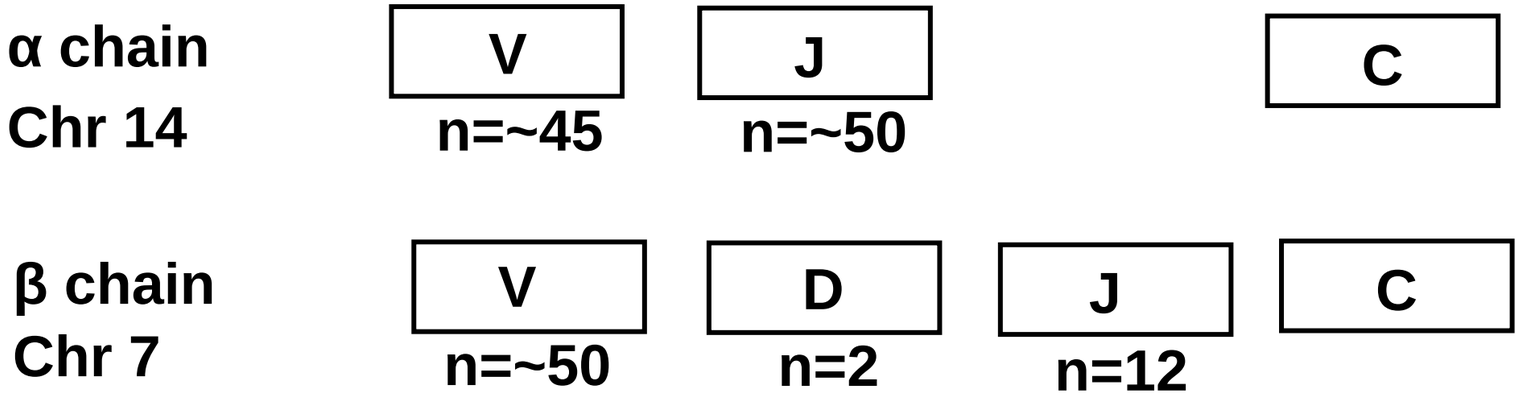
CDR1 and CDR2 encoded by germline in V



CDR3 by DJ and junctional for VDJ B and J + VJ junctional for a

junctional diversity
(non-germline)
largest contributor to diversity

combinatorial diversity
(germline)
1 to 3 million upper limit)

P nucleotides (template after hairpin opening by Artemis)

N nucleotides (then nontemplate followed by DNA ligation) (20 by TdT) - more common

10^{15} - 10^{20} unique TCR combhinations